

RAJ DESAI

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SUMMARY

Full-Stack Engineer with 2 years of hands-on experience building end-to-end applications, from React frontends to Node.js backends with PostgreSQL, Redis, and async pipelines. Specializing in API design, real-time systems, and LLM integration. Currently completing a master's in computer science at UT Dallas, bringing a build-first, figure-it-out approach to every problem.

EDUCATION

The University of Texas at Dallas | MS – Computer Science Dallas, TX | Expected May 2026
Relevant Courses: Design Analysis of Algorithms, Machine Learning, NLP, AI, Big Data Analytics, Database Design.
University of Mumbai | BE – Computer Engineering Mumbai, India | May 2024

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, Dart, C/C++, SQL, HTML, CSS
Frontend: React, Next.js, Tailwind CSS, Flutter, Framer Motion, Recharts
Backend: Node.js, Express.js, FastAPI, REST APIs, Redis, BullMQ, OAuth 2.0, JWT
Databases: PostgreSQL, MySQL, MongoDB, Prisma ORM
ML and AI: LLMs (OpenAI, Gemini APIs), FAISS, Sentence Transformers, Pandas, NumPy
Tools & Platforms: Git, GitHub Actions, Docker, CI/CD, Vitest, Vercel

PROFESSIONAL EXPERIENCE

Software Engineer Intern - Hydracoral Technologies Jan 2023 – Dec 2023 | Mumbai, India

- Accelerated user verification by 30% by developing a Flutter/Dart onboarding flow with facial-recognition APIs, validated in staging
- Reduced app crash rate by 20% by diagnosing and resolving 15+ bugs in state management, API error handling, and memory leaks across the Flutter codebase using Firebase Crashlytics and structured debugging workflows
- Improved usability scores by 20% by designing responsive cross-platform UI components validated through structured user-testing
- Owned CI/CD pipelines in Git and Bitbucket for a 3-person Agile team, integrating REST APIs for JWT authentication, data sync, and release automation across mobile and backend services

PROJECTS

Yoru - Cinematic Ambient Music Streaming Web App Jan 2026 – Feb 2026

- Shipped a cinematic streaming app with Next.js 15, TypeScript (strict), and Tailwind CSS, orchestrating 21 scenes with Framer Motion crossfades and dual-channel Howler.js audio mixing
- Engineered a component-driven codebase with custom React hooks for idle detection and keyboard navigation, achieving WCAG 2.1 AA compliance and >80% test coverage via Vitest

PR Sensei - AI-Powered GitHub Code Review App Nov 2025 – Jan 2026

- Automated pull-request reviews to under 30 seconds by developing an AI GitHub App with a React frontend and Node.js backend that posts summaries and inline comments via OpenAI/Gemini APIs, triggered by webhook events
- Solved GitHub's 10s webhook timeout against 5–15s LLM calls by engineering an async pipeline with Redis and BullMQ, handling retries, rate limiting, and graceful degradation
- Centralized review analytics and file-hotspot tracking across repos by designing a multi-tenant PostgreSQL schema with Prisma ORM, served to a React dashboard with Recharts visualizations and per-repository filtering

LLM Cookbook - AI-Based Semantic Recipe Search Mar 2025 – Apr 2025

- Delivered sub-1-second semantic search over 10,000+ recipes using FAISS-indexed MiniLM embeddings served through a full-stack app (Streamlit frontend + FastAPI backend), validated across 200 test queries
- Enabled natural-language recipe discovery by designing RESTful API endpoints with multi-dimension filtering by diet, allergens, and cuisine, powered by Sentence Transformers with ranked relevance scoring
- Processed and normalized 10,000+ raw recipe records into model-ready embeddings via a Pandas pipeline for extraction, cleaning, and vectorization, reducing manual data-prep to a single reproducible script

Financial SaaS Platform - Real-Time Banking Dashboard Oct 2024 – Dec 2024

- Unified multi-account financial data into a single real-time dashboard with Next.js, React, TypeScript, and Tailwind CSS, displaying transaction feeds across checking, savings, and credit accounts
- Connected live banking data from multiple institutions by integrating Plaid API for aggregation and Appwrite for real-time updates, securing access with OAuth 2.0 authentication, session management, and token refresh